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Table 13.1 Different types of Silkworm

Species of silkmoth	Silk Producing States	Preferred Food (Leaves)	Type Of Silk
Bombyx mori	Karnataka, Andhra Pradesh and Tamil Nadu	Mulberry	Mulberry Silk
Antheraea assamensis	Assam, Meghalaya. Nagaland, Arunachala Pradesh and Manipur	Champa	Muga Silk
Antheraea mylitta	West Bengal, Bihar and Jharkand	Arjun	Tassar Silk
Attacus ricini	Assam, Meghalaya. Nagaland, Arunachala Pradesh and Manipur	Castor	Eri Silk

feed during its very short life period of 2-3 days. Just after emergence, male moth copulates with female for about 2-3 hours and if not separated, they may die after few hours of copulating with female. Just after copulation, female starts egg laying which is completed in1-24 hours. A single female moth lays 400 to 500 eggs depending upon the climatic conditions. Two types of eggs are generally found namely diapause type and non-diapause type. The diapause type is laid by silkworms inhabiting the temperate regions, whereas silkworms belonging to subtropical regions like India lay non-diapause type of eggs. The eggs after ten days of incubation hatch into larva called as caterpillar. The newly hatched caterpillar is about 3 mm in length and is pale, yellowish-white in colour. The caterpillars are provided with well developed mandibulate type of mouth-parts adapted to feed easily on the mulberry leaves.

After 1st, 2nd, 3rd and 4th moultings caterpillars get transformed into 2nd, 3rd, 4th and 5th instars respectively (Figure13.3). It takes about 21 to 25 days after hatching. The fully grown caterpillar is 7.5 cm in length. It develops salivary glands, stops feeding and undergoes pupation. The caterpillars stop feeding and move towards the corner among the leaves and secretes a sticky fluid through their silk gland. The secreted fluid comes out through spinneret (a narrow pore situated on the hypopharynx) and takes the form of long fine thread of silk which hardens on exposure to air and is wrapped around the body of caterpillar in the forms of a covering called as cocoon. It is the white coloured bed of the pupa whose outer threads are irregular while the inner threads are regular. The length of continuous thread secreted by a caterpillar for the formation of cocoon is about 1000-1200 metres which requires 3 days to complete. The pupal period lasts for 10 to 12 days and the pupae cut through the cocoon and emerge into adult moth.

On the basis of the moults which they undergo during their larval life, *B. mori*